

Will Jameson

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will-jameson.github.io/portfolio-website

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PROFESSIONAL SUMMARY

Electrical Engineering graduate (2026) with hands-on experience in power electronics, embedded systems, and PCB design. Skilled at taking projects from concept and schematic through to working prototype. Background in fast-paced hospitality environments has sharpened communication, composure under pressure, and attention to detail.

TECHNICAL SKILLS

Microcontrollers & FPGAs: STM32, FPGA (Verilog, Efinix, ModelSim, Quartus, GTKWave)

EDA & Simulation: KiCad, EAGLE, Fusion 360, LTSpice, Simulink, SolidWorks

Programming: C, C++, Python, MATLAB, BASH

Circuit Design: Power electronics, op-amp circuits, buck/boost converters, PCB layout

Protocols & Interfaces: I2C, UART, SPI, PWM

Test Equipment: Oscilloscopes, DMMs, function generators, logic analyzers

PROJECT EXPERIENCE

Buck Converter Design - PCB & Power Electronics

- Designed a 5V - 1.8V synchronous buck converter around the TI-LM21215 IC, achieving < 30% inductor ripple current and < 2% output voltage ripple - meeting all design targets on first board spin
- Laid out complete PCB in Autodesk EAGLE, applying best practices for switching power supply layout: tight high-current loops, proper ground planes, and component placement to minimize EMI
- Validated performance with oscilloscope measurements across load conditions; iterated on compensation network to improve transient response

Tremor-Correcting Drafting Robot - Robotics, Embedded, and Digital Signal Processing

- Designed an assistive Cartesian plotting robot to help artists with essential tremor draw and write with precision, using an FIR digital filter to suppress involuntary hand-motion frequencies in real time
- Integrated SPAD time-of-flight sensors for accurate X-Y position sensing and implemented digital filtering pipeline on STM32 microcontroller
- Built full mechanical and electrical system: integrating stepper motor drive electronics, motion control firmware, and pen-lift servo - successfully demonstrated smooth, tremor-free output from filtered inputs
- reclaimcreative.github.io/stabilize

Automated Louvers for Passive Cooling - Power & Embedded Systems

- Designed and built a solar-powered louvre control system to automate passive building ventilation, reducing reliance on active HVAC

- Implemented control loop on embedded microcontroller to drive louver actuators and maintain target louver angle based on temperature feedback
- Designed power conditioning circuitry to source current from solar cell array or main power supply; managed I2C communication between sensors and control MCU

WORK EXPERIENCE

Teacher's Assistant - Introductory Circuit Analysis - 2023

City College of San Francisco Engineering Department - San Francisco, CA

- Supported 30+ students in understanding foundational circuit analysis concepts, including Kirchhoff's laws, Thevenin/Norton equivalents, and AC analysis
- Led weekly review sessions, designed supplemental practice problems, and stood in as needed to deliver full lessons in the instructor's absence
- Provided one-on-one coaching during office hours, consistently earning positive feedback for clear and patient explanations

Hospitality Industry

Casement's Bar - Bartender - San Francisco, CA - 2020 - present

- Delivered consistent, high-quality service in high-volume environments, professionally managing competing priorities
- Developed strong cross-functional communication skills working alongside kitchen and FOH teams to resolve guest issues quickly and professionally
- Maintained meticulous accuracy in order management, billing, and cash handling across hundreds of transactions per shift
- Trained and mentored new staff on service standards, product knowledge, and floor procedures - reducing onboarding time and improving team consistency

EDUCATION

B.S. Electrical Engineering - 2026

San Francisco State University - San Francisco, CA

A.S. Engineering - 2024

City College of San Francisco - San Francisco, CA

- Teacher's Assistant for Introductory Circuit Analysis; proficient with oscilloscopes, DMMs, and standard bench test equipment

B.A. Chinese Language - 2009

San Francisco State University - Study Abroad Beijing University (1 year) - San Francisco, CA

- Assisted SFSU Chinese Department with document translation; professional working proficiency in Mandarin